Marcus Rocco Fratarcangeli

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PROFESSIONAL SUMMARY

- Mechanical engineer (M.S., Georgia Tech) specializing in engineering automation, intelligent systems, and software enabled-laboratory infrastructure
- Experienced in developing integrated hardware and software solutions spanning robotics, computer vision, environmental testing, and machine learning to improve engineering efficiency, quality, and decision-making

SKILLS

Software: SolidWorks, Python, VBA, Creo, MATLAB, Arduino, Machine Learning, Computer Vision, GitHub, Microsoft Access, Microsoft Excel, Microsoft Word, Microsoft PowerPoint
Technical: Additive Manufacturing (AM), Electromechanical Systems, Robot Programming, Rapid Prototyping, System Integration, Sensors, Automation, Design for Manufacturing (DFM), Metrology, Measurement Systems
Interpersonal: Project Management, Leadership, Public Speaking, Technical Presentations

EXPERIENCE

L3Harris — Palm Bay, FL
Senior Associate, Integration & Test Engineering — January 2026 – Present

- Engineered a configurable VBA-based automation platform that generates standardized environmental test procedures from engineering inputs, reducing document preparation time by 97% and saving over $500k annually
- Designed a scalable rule-based workflow using extensive conditional logic to accommodate hundreds of environmental test variations while enforcing standardized language, formatting, and procedural sequencing to reduce documentation errors and nonconformances
- Executed thermal, thermal-vacuum, and vibration testing of spaceflight hardware, ensuring compliance with program requirements while supporting safe operation of high-value aerospace systems
- Developed an automated OCR-based workflow that extracts engineering documents from email, processes scanned records, and organizes documentation into structured folders, reducing manual administrative effort
- Designed an internal training and certification management tool used by engineers and administrators to streamline tracking of required qualifications and improve workforce readiness
- Collaborated with cross-functional engineering and technician teams to troubleshoot test issues, support environmental test execution, and improve laboratory operations

Active Materials and Additive Manufacturing (AM^2) Lab — Atlanta, GA
Graduate Research Assistant — August 2023 – August 2025

- Developed and built a fully autonomous tensile testing system with robotic arm integration, custom clamping, digital image correlation, and Python-based ML anomaly detection, enabling increased testing throughput to 48 samples in under four hours, reducing the human time by over 88%
- Established an autonomous curing kinetics system to evaluate resin viability for DLP 3D printing, cutting manual analysis time from hours to minutes, and enabling rapid down-selection of promising photopolymers
- Designed and assembled a high-throughput photopolymer composition system enabled by a robotic arm, bespoke closed loop liquid dispensing system, and a custom powder dispensing device which can automatically create and mix photopolymer inks without human intervention
- Integrated testing systems with automated data analysis pipelines on GitHub, creating a self-driving lab platform that reduced the overall human time by 91.3%, with an automation factor of 11.4x, and enabled machine learning models to guide experimental design in real-time

SELECTED PROJECTS

***true* Rate of Perceived Exertion (*t*RPE)** Huntington, NY
RPE Predictive Model for Powerlifting August 2025 – Present

- Addressing a gap in performance tracking for the ~100,000 active U.S. powerlifting athletes who rely on RPE to measure training effort, with the goal of reducing uncertainty and enabling more evidence-based training
- Developed a Python/OpenCV computer vision pipeline with ArUco detection to track 3D barbell motion in SBD videos, extracting per-rep metrics, such as peak velocity, range of motion, and time under tension
- Building an application that records and processes athlete videos, delivering real-time feedback and generating training data for machine learning-based RPE prediction models

QUbeSat (Senior Design Capstone) Hamden, CT
Mechanical Engineering Senior Design May 2022 – May 2023

- Led a team of three senior engineering students to design and build a CubeSat from concept to working model utilizing a bespoke 3D printed frame with an integrated thermal sensing apparatus
- Developed a project plan, assigned roles, conducted research, and coordinated with outside experts
- Presented a functioning prototype at the ASEE Zone 1 Conference and received an Award for Excellence

EDUCATION

Georgia Institute of Technology Atlanta, GA
Master of Science in Mechanical Engineering August 2023 – August 2025
Major area: Manufacturing, Minor area: CAE/Design
Cumulative GPA: 3.71/4.0
Advisor: Dr. H. Jerry Qi
Thesis topic: "Self-Driving Lab of Photopolymer Synthesis and Mechanical Testing for Expedited Material Discovery"

Quinnipiac University Hamden, CT
Bachelor of Science in Mechanical Engineering August 2019 – May 2023
Minors: Entrepreneurship and Mathematics
Cumulative GPA: 3.95/4.0, Engineering GPA: 4.0/4.0
Honors and Awards: Outstanding Achievement in Mechanical Engineering (2023), Dean's List (2019-2023)

SELECTED PUBLICATIONS

Gholami, F., Yue, L., Li, M., Jain, A., Mahmood, A., **Fratarcangeli, M.**, ... & Qi, H. J. (2024). Fast and efficient fabrication of functional electronic devices through grayscale digital light processing 3D printing. *Advanced Materials, 36*(46), 2408774.

Li, M., Gholami, F., Yue, L., **Fratarcangeli, M.** R., Black, E., Shimokawa, S., ... & Qi, H. J. (2024). Coaxial-Spun Hollow Liquid Crystal Elastomer Fiber as a Versatile Platform for Functional Composites. *Advanced Functional Materials, 34*(42), 2406847.

CERTIFICATIONS

NCEES Fundamentals of Engineering (FE Exam Passed) May 2024
Dassault Systèmes SOLIDWORKS Associate Mechanical Design May 2023

LEADERSHIP

Mentor, *Georgia Tech ENGAGES Program* May 2024 – May 2025

- Mentored an Atlanta high school student in a subset research project developing an automated algorithm for analyzing tensile data which was presented at the Georgia Science and Engineering Fair (GSEF)

Vice President, *Eta Pi Engineering Honor Society* August 2022 – May 2023
Vice President, *Quinnipiac Club Powerlifting Team* August 2022 – May 2023
Eagle Scout, *Boy Scouts of America* May 2019 – Present